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SURVEY

## A Survey of Explainable Artificial Intelligence (XAI) in Financial Time Series Forecasting

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# A Survey of Explainable Artificial Intelligence (XAI) in Financial Time Series Forecasting

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Artificial intelligence (AI) models have reached a very significant level of accuracy. While their superior performance offers considerable benefits, their inherent complexity often decreases human trust, which slows their application in high-risk decision-making domains, such as finance. The field of explainable AI (XAI) seeks to bridge this gap, aiming to make AI models more understandable. This survey, focusing on published work from 2018 to 2024, categorizes XAI approaches that predict financial time series. In this article, explainability and interpretability are distinguished, emphasizing the need to treat these concepts separately, as they are not applied the same way in practice. Through clear definitions, a rigorous taxonomy of XAI approaches, a complementary characterization, and examples of XAI's application in the finance industry, this article provides a comprehensive view of XAI's current role in finance. It can also serve as a guide for selecting the most appropriate XAI approach for future applications.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Applied computing** → *Economics*; • **Computing methodologies** → *Machine learning*; Artificial intelligence;

Additional Key Words and Phrases: Explainable artificial intelligence, XAI, interpretable model, explainable model, time series, finance

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## 1 Introduction

Since the mid-2010s, **artificial intelligence (AI)** has made remarkable strides in terms of accuracy, transforming sectors such as business, media, healthcare, education, finance, scientific research, judicial, and so on. While these advances offer unprecedented capabilities, they also introduce challenges, particularly in high-stakes, risk-sensitive domains like finance and healthcare. Despite their accuracy, AI models often suffer from a lack of user trust due to their black-box nature. Ethical and legal frameworks have evolved to reflect these concerns; reports such as References [77] and [95] highlight the necessity for AI systems to be understandable in decision-making contexts. As

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pointed out by existing works [10, 69, 77], the poor interpretability of deep learning models can significantly increase investment risks, raising critical limitations for practical applications.

Addressing this problem is the emerging field of **Explainable Artificial Intelligence (XAI)**, designed to produce AI models that are not only accurate but also understandable to human users. XAI offers two primary avenues for enhancing understanding: interpretable models, which are inherently understandable, and explainable models, which are models that require additional methods for explanation. However, there exists significant terminological confusion in the literature; the terms interpretable and explainable are often used interchangeably, leading to misunderstandings about the nature of the model being presented. Understanding the differences between these terms is critical for several reasons. First, the pathways to achieve interpretability and explainability are different, necessitating distinct methodologies and evaluations. Second, accurate terminology helps in the selection of appropriate models based on the unique regulatory and ethical demands of financial applications. Third, clear definitions improve interdisciplinary communication among data scientists, financial experts, and policymakers, facilitating more informed decisions. Fourth, distinguishing between interpretability and explainability informs the level of trust one can place in a model's decision-making process, impacting risk assessment and compliance with financial regulations. Furthermore, the claim that a model is XAI sometimes lacks substantiation, either by failing to show the interpretations that could be done on the model or by lack of the explanations that help understand the model.

The need for this survey is further amplified by the financial industry's accelerating adoption of AI for complex tasks such as credit risk assessments, fraud detection, algorithmic trading, and, notably, the prediction of financial time series. Although these technologies promise to enhance efficiency and accuracy, they also raise significant concerns around fairness, explainability, and regulatory compliance. The opaque nature of AI algorithms is not merely a technological challenge; it has far-reaching implications for public trust, ethical governance, and regulatory adherence. This article aims to present the latest interpretable models and explainability methods used in the prediction of financial time series. By offering a detailed survey of both interpretable and explainable models, it aims to serve as a guide for two groups of readers: data scientists who want to design and develop XAI into time-series prediction, whether financial or not, and financial professionals who want to incorporate XAI into their analytics environment for predicting financial time series. Indeed, the interpretable and explainable approaches surveyed in this article for predicting financial time series could also be used to forecast many types of time series. Also, this survey provides financial insights that the XAI made possible. Both the insights and their approaches can be useful for financial professionals. Additionally, this survey provides a rigorous foundation for continued investigation, guides practitioners in choosing the most appropriate approach, and informs them about the merits and limitations of various approaches. In essence, this survey aspires to facilitate the responsible and transparent application of AI in finance, aligning cutting-edge machine learning techniques with the sector's strict ethical and regulatory standards. To the best of our knowledge, this is the first survey focusing specifically on XAI for predicting financial time series.

In this article, we categorize XAI models from 2018 to 2024 that have been used to predict financial time series or to predict financial time-series movement. First, some important concepts are defined in the Section 2. Second, the methodology of this survey and the diagram resulting from the taxonomy are presented (Sections 3 and 4). Third, interpretable models and explainability methods are presented and classified according to its XAI principle (Sections 5 and 6). In Section 7, the same approaches are characterized according to different criteria, namely the explanation characteristics known by the community [6, 26, 55]. Next, applications and the challenges of the XAI are presented (Section 8). A comparison of XAI with other fields is then discussed (Section 9). Finally, conclusions and the work that remains to be done are presented (Section 10).

## 2 Definitions

The financial industry has increasingly adopted AI, since the inception of many statistical models in the 1980s, primarily for stock trading and risk management applications. As the domain matured, it embraced more advanced techniques, from expert systems and neural networks to the recent inclusion of deep learning models. These innovations have provided the industry with remarkable tools for market analysis, financial behavior prediction, and risk management. However, the lack of transparency in these complex models often limits their practical application. When users cannot decipher a model's reasoning, even high-accuracy outputs might be deemed unreliable for critical financial decisions. This emphasizes the vital role of XAI as a solution.

The XAI is a branch of AI designed to be comprehensible to humans, thus promoting trust among users. XAI's main feature is its ability to render the inner workings of its model intelligible to human users. Its importance in finance manifests in various ways. For instance, when a financial institution deploys AI for decision-making regarding a client, as in the case of a loan application, there is a pressing need to clarify the rationale behind such decisions. This transparency is imperative not only for individual clients but also for communicating decision-making processes to investors, board members, or auditors. XAI facilitates this clear communication. Additionally, when institutions rely on AI for critical decisions, stakeholders require insights into how those decisions are derived, ensuring their sustained trust in the system. From a risk management perspective, an opaque AI model introduces potential hazards. Comprehending the model's operations enables institutions to identify and mitigate risks more effectively. In decision-making scenarios, XAI provides pertinent information, helping users gauge the trustworthiness of predictions. Thus, users can judiciously combine the model outputs with their domain expertise to arrive at well-informed decisions.

To qualify as XAI, an AI system must either be inherently interpretable (interpretable model) or use a distinct method that clarifies its decision-making processes (explainable model). An *interpretable model*, often referred to as a transparent model, is designed in such a way that users can intuitively understand its inner workings just by examining it. The individual components of the model are clear and understandable, facilitating a comprehensive grasp of its functionality. As an example, a linear model with a limited set of input features is considered interpretable. In such a model, each coefficient indicates the contribution of its associated feature. Thus, *interpretability* can be characterized as the inherent property of a model that, by its very design, makes its operations transparent at first sight. Interpretability is not a simple binary trait and can be evaluated by some criteria presented in Section 5.4.

An *explainable model* is designed to provide explanations to users, aiding their comprehension of the model's operations. Often, an explainable model is essentially a black-box model enhanced with specific explainability methods, such as **Shapley additive explanations (SHAP)** [66] or **Local Interpretable Model-agnostic Explanations (LIME)** [91], to shed light on its internal processing. A pure *black-box model*, also known as an opaque model, operates in a manner that is not immediately discernible, making its internal workings and decision-making processes inscrutable to users. This lack of transparency is typically due to the model's inherently complex design but can also arise from confidentiality concerns. *Explainability* can thus be defined as the capability of a model to elucidate its functioning, offering users insights into its operational mechanics.

It is essential to define a few basic concepts that are frequently used in the rest of the article. The term *model* refers to a series of calculations, or mathematical equations, describing the workings of a computing system that takes typical inputs, such as numbers or words, and generates outputs, or predictions, for a given task. An *explainability method* is defined as a process that takes a model as input and provides explanations of this model as output. Consequently, an explainable model is

one on which an explainability method has been applied. An *approach* encompasses a number of these computation steps. We can define it as a process that takes an input and generates an output.

As highlighted in Reference [82], ambiguity persists regarding the fundamental concepts of XAI, particularly in defining the target audience and the nature of explanations. Two key issues stand out as follows:

- (1) The term *user* remains undefined. In finance, it could refer to data scientists, business professionals, auditors, or consumers, each requiring tailored explanations. Specifying the audience is therefore critical for rendering a model explainable or interpretable [7, 9, 71, 82].
- (2) The notion of *explanations* lacks a formal definition and must adapt to user-specific contexts [8, 9]. For instance, financial time-series prediction often demands insights into feature importance, decision rules, and time-series patterns. Visual aids, such as dashboards, further enhance understanding by integrating explanations with supplementary information. These aspects—feature importance, decision rules, trends interpretation, and visual explanations—form the *XAI principles* underpinning the approaches discussed in this article.

Let us apply these definitions. Portfolio managers using the output of a model to make investment decisions tend to focus on trusting or understanding the forecasting rather than on extensive explanations. They may benefit from a confidence score and local feature importance, which highlights inputs relevant to a specific prediction. This helps them assess whether a decision is well founded or flawed. In contrast, model developers might be more interested in global importance of feature, because the global explanations pertain to the training of a model. Both local and global explanations complement each other for a deeper understanding of the model's behavior. However, when features lack financial meaning or are present in large number, their importance may be difficult to interpret. Grouping features into financial themes (e.g., volatility, momentum) may offers a more actionable approach for portfolio managers.

The previous discussions suggest that the explanations should be adapted to the context and the user for optimal effectiveness. However, the evaluation of the explanations should be objective, i.e., based on rigorous criteria. Adjustment to the context could be one of these criteria, but the evaluation itself should not depend on the subjectivity of the user. We present some ideas of criteria in Section 6.3.

### 3 Methodology

In this article, we categorize approaches that are either explainable or interpretable and used for predicting financial time series or their movement. These time series may include stock prices, volatility, and macroeconomic indicators, among others. Our research centers on four key concepts (XAI, time-series forecasting, machine learning, and finance), and we based our search criteria on these. Specifically, we identified relevant articles using keywords such as “explainability,” “finance,” “forecasting,” and “model.” Alternative terms used for “explainability” could be “interprétabilité,” “interpretability,” “XAI,” “explainable,” “interpretable,” “explicability,” “feature importance,” or “transparent model.” “finance” substitutes include “stocks,” “volatility,” “tendency,” “financial,” and “market prices.” “forecasting” has been replaced by “time-series forecasting,” “regime,” “regime classification,” and “regime clustering.” Similarly, “model” have been replaced by “AI,” “method,” “machine learning,” or “artificial intelligence.” This inclusive approach aimed to maximize coverage without excluding relevant studies, focusing on our core concepts. However, additional terms like “decision rules,” “visual explanation,” or “dashboard” might have enriched the search further. We conducted a comprehensive search in various databases, including SCOPUS, Inspec, IEEE Xplore, Computers and Applied Sciences Complete, ACM Digital Library, and arXiv. In the initial submission of this paper, we restricted our search to papers published between 2018 and June 2023, a

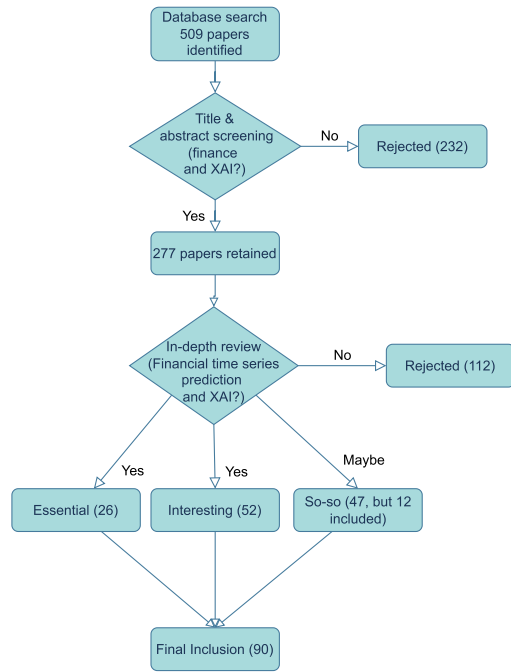


Fig. 1. Article screening and selection.

period characterized by a significant increase in publications on XAI, particularly in the financial domain. In total, we identified 509 papers.

As indicated in Figure 1, from the 509 identified papers, we applied a two-step filtering process. First, we reviewed the titles and abstracts, retaining only papers that explicitly addressed both finance and XAI. We excluded those that focused on XAI but were unrelated to finance, such as applications in ecology, as well as finance-related papers that did not incorporate XAI or machine learning. This step reduced the dataset to 277 papers. In the second phase, we examined key sections of these papers to extract relevant information on XAI and its financial applications. Based on their relevance, we classified them into four categories: essential (26 papers), interesting (52 papers), so-so (47 papers), and rejected (112 papers). The essential and interesting papers were fully included in our review, while approximately 12 of the so-so papers were considered. None of the rejected papers were retained. Since our study focuses on predicting financial time series or their trends using explainable or interpretable models, we included any paper that aligned with this scope. Otherwise, we excluded it. For instance, we discarded papers on models for credit risk assessment and loan approval prediction, as these models forecasted discrete values or classes independent of time rather than time-series data. After applying these criteria, 90 papers were retained for detailed analysis.

Our emphasis is on the interpretability and explainability aspects of the models discussed. The aim of this article is to provide an overview of advancements made in the explainability and interpretability of models designed for predicting financial time series. Some models use time-series data as input, others rely on text from news articles or the internet, and some utilize both. It is worth noting that the rising popularity of XAI has led to its use as a buzzword in some cases, complicating the classification process. While the primary focus was on papers derived from our systematic methodology, additional sources were included to provide context and enrich

the discussion of the presented information. It is true that some articles received more attention than others in this review; however, this should not be interpreted as an indication of their relative importance. We may also have inadvertently omitted articles that merit being included. Indeed, our review process involved reading specific sections of the papers to focus on the interpretability and explainability of the models. While our aim was to conduct as a thorough search as possible, there is a chance that we may have missed some relevant articles on the topic.

During the 2024 revision process, we conducted a supplementary review using the same methodology described above, focusing on papers published between 2023 and 2024. The goal of this additional research was to incorporate the latest trends and developments on the topic. While this review was carried out with slightly less rigor than the original systematic approach, it allowed us to introduce a sense of novelty to the article, as the initial research had been conducted some time ago.

#### 4 Taxonomy

The goal of the taxonomy presented in the Figure 2 is to help readers identify the right XAI approach for their specific context. Initially, XAI approaches are divided into interpretable models and explainable methods. Each category is then further sorted based on the principles of XAI, including feature importance, visual explanations, and trend. After that, explainability methods and interpretable models are categorized based on the technique used to illustrate the XAI principles. For the interpretable models, it includes linear regressions, attention mechanisms, decision trees and other techniques. For explanation methods, it includes perturbation, propagation, and visual interface. The reviewed approaches are presented in Sections 5 and 6 following the diagram presented in Figure 2.

#### 5 Interpretable Models

This section presents the interpretable models reviewed in the survey. These models provide feature importance (see Section 5.1), decision rules (refer to Section 5.2), or analysis of relations like trends in the time series (as discussed in Section 5.3). The evaluation of the models' interpretability is discussed in Section 5.4.

Some of the presented models may appear to be more like black-box models than interpretable models. These models, because of their complexity, do not fit the conventional definition of interpretable models that only regroups basic and simple models. For instance, neural networks, attention models and other complex models are usually considered as black-box models by the community. This makes sense, because they are complex compared to basic interpretable models such as linear models and decision trees. Based on that criterion, complex models, like the attention models, would have been excluded from the class of interpretable models. However, the XAI models are not classified according to their complexity. To be considered as interpretable, the models need to provide some information that helps the user understand them, specifically through XAI principles. Therefore, the origin of the XAI principles is the key factor that determines the nature of a model with regard to interpretability and explainability. If the principle directly comes from the model itself, then the model is considered interpretable. Otherwise, if the XAI principle comes from another algorithm, namely an explainable method, then the model will be considered as an explainable model. So, even if some models are complex, they can still generate information that can be interpreted by the user to better understand the process behind the model. This is why many authors, including us, refer to them as interpretable. For example, attention models, like the ones presented in Section 5.1.2, contain attention weights that represent the importance of the features. As these weights come from the models themselves, they are classified as interpretable models. It is also the case with complex neural networks that are classified as interpretable due to

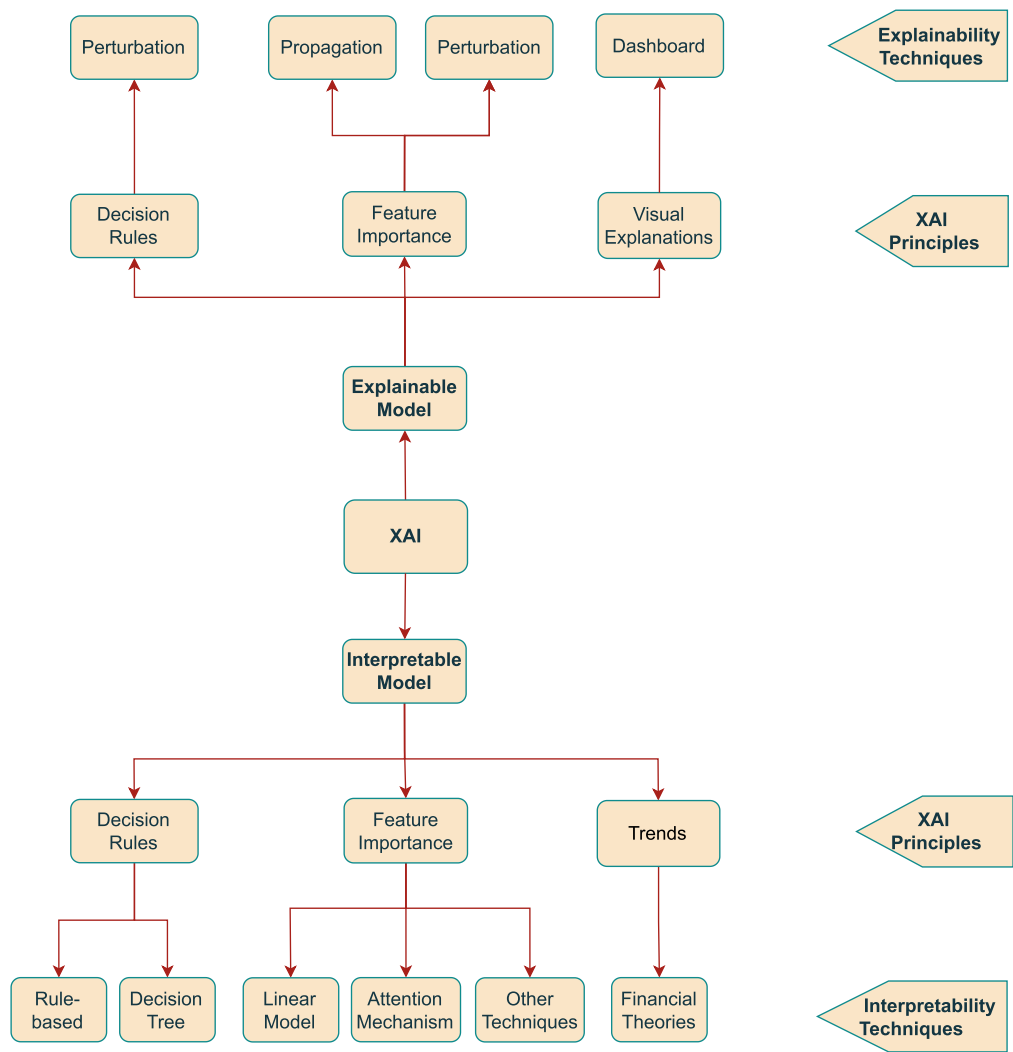


Fig. 2. Classification of XAI approaches.

a component in their structure, like a mask [97] or components of fuzzy logic [108], that reveals some information about the networks to users.

There may exist other definitions of interpretability and explainability that could lead to a different classification of complex models. However, we believe that the definitions provided in this article help draw a clear distinction between interpretability and explainability. Indeed, even if some models seem to fit into both categories, the implications of these definitions establish a clear practical distinction between interpretability and explainability. In interpretable models, where XAI principles are intrinsically embedded within the prediction process, altering or removing these principles directly affects the model's predictions. In contrast, explainable models derive their XAI principles from external explainability methods, rather than from within the prediction mechanism itself. As such, these XAI principles can be removed without impacting the model's outputs. When the classification of a model is ambiguous, this distinction serves as a secondary criterion,

whether removing the XAI principles alters the prediction or not. As a result, the classification of these models based on these definitions is both logical and easy to understand.

## 5.1 Feature and Time Importance

*Feature importance* is an XAI principle that measures how much each feature has an effect on the prediction or on an entire dataset. It also includes time importance, which measures how much specific time periods were important. If the feature importance reflects the effect on a specific prediction, then it is called local feature importance. If it reflects the effects on a dataset, then it is called global feature importance. Usually, the global feature importance is computed on the training dataset, but it can also be computed on the test set. By ranking the features according to their importance, the important features can be extracted. The feature importance can be plotted through heat maps, bar plot, line plot, and so on. The feature importance is computed for all types of features, including both input features and internal features, which represent learned representations within the model. It is important in finance, because it shows which information the model uses to make its predictions. The comparison between financial analysts and a model can easily be made. In their work, the analysts will, unconsciously or consciously, weight the information that they will read and make a prediction based on the important one. A model, showing the importance of its features, will give the same kind of information to the user.

Feature importance brings both challenges and opportunities. For instance, having a high number of features can obscure the understanding of each feature's importance. To bypass this issue without altering the model itself, one strategy is to aggregate features into clusters and then assess their importance at the cluster level. If some features do not have a financial meaning, then it becomes more challenging to interpret the results. However, when they do, it is possible to draw more detailed conclusions about interpretability. This connection between the model and the context enables users to better understand both. When the context is well known, feature importance can help justify the model's decisions. Furthermore, the feature importance of an accurate model can reveal unknown information about the financial market and help users develop new theories. Feature importance can be extremely useful for financial analysts or portfolio managers for several reasons. For instance, it helps identify key drivers of market behavior, especially when the features are meaningful and interpretable, bridging the gap between quantitative analysis and fundamental analysis. At a local level, since financial markets are often volatile and unpredictable, feature importance clearly indicates which variables can drive significant market movements at a given moment. For example, if an increase in trading volume strongly signals a market correction, then feature importance can highlight this relationship, offering valuable insights for decision making. Additionally, feature importance can be used to eliminate irrelevant or noisy features, thereby improving the model's performance and robustness. Moreover, feature importance evolves over time, reflecting changes in market dynamics. These shifts in feature relevance can be used to characterize specific market periods, helping analysts adapt their strategies to align with the prevailing market conditions.

In this article, various techniques to compute feature importance are discussed. These include coefficients in a linear model, gain in decision trees, relevance scores, and attention weights. It is important to note that while all these techniques aim to measure feature importance, they might not all convey the same underlying concept. Therefore, it will be important to evaluate the differences between these approaches in the future.

**5.1.1 Linear Regressions.** The first model that comes to mind when discussing feature importance is the linear model. It is one of the most interpretable models due to its simple way of learning and the feature importance that it provides. However, a vanilla linear model cannot

precisely predict financial time series due to their complexity and nonlinearity. For this reason, the vanilla linear model is not used in practice for predicting financial time series. However, linear regression can be integrated into a complex model that has the ability to predict financial time series, enhancing its interpretability. We define the linear regression  $f(x)$  as  $f(x) = \beta_0 + \sum_j \beta_j x_j$ , where  $x_j$  are the features and  $\beta_j$  are the coefficients of the associated feature  $x_j$ . These coefficients represent the effect of their associated features. The features can be the input ones [74, 120], but they can also be internal features [127].

A common way to use a linear regression in a complex model is to compute the weights of the regression with a complex model. This way allows a tradeoff between interpretability and accuracy: It loses in the simplicity of the algorithm to learn the weights, but it gains in accuracy. For example, to predict unexpected revenues for companies, an Adaptive Master-Slave model was proposed in Reference [120]. The model has different weights for each company, unlike the normal linear models that would have the same weights for all companies. Thus, each company has their important and nonimportant features that help the user to understand more locally the predictions. As its name suggests, the model is composed of a master model and a slave model, with the latter being a linear model responsible for predicting the output. Munkhdalai et al. [2022] proposed also an architecture composed of a linear model and a complex model. In that case, the complex model is a recurrent neural network. Indeed, to predict time series, including the price of the Nasdaq, from temporal numerical data, the authors [74] use an adaptive linear regression, where the weights are dynamically adjusted over time based on the recurrent network. The prediction model is therefore interpretable locally by the coefficients of the linear regression. The average of the coefficients were done to compute the global importance. For example, the global importance of the “DTWEXB\_1,” which is the 1-day lag of trade weighted U.S. dollar index is  $-0.58$ . It means that an increase of one point of this feature would cause a drop of a 0.58 points of the Nasdaq index [74]. Since the coefficients of the model change through times, they were plotted as time series. These time series can be analyzed with classic statistics like the trend, the volatility, and so on. For instance, it shows that the importance of “DTWEXB\_1” was highly volatile between 2015 and 2016. It also shows that “DTWEXB\_1” was more important in 2017 than before.

A different way of integrating linear regression into a complex model involves projecting the input into meaningful dimensions and then applying a linear combination of these dimensions with the appropriate weights. This is the concept behind the semantic path model described by [36], where text is used as the input. Indeed, this model helped the authors to better understand the economic context of the Europe. More specifically, it projects the input words into a reduced latent semantic space and forecasts macroeconomic indicators. The dimensions of the latent space are interpretable and are represented by types of words such as “negative outlook,” “economic uncertainty,” “legal risk,” and so on. The importance of the dimensions of the latent space was analyzed and some financial conclusion has been made. For example, the authors found that the negative dimension, that represents negative news, affects many indicators, namely the unemployment rate, inflation, industrial production, consumer confidence, and government bonds. Similarly, uncertainty and legal constructs were found to have distinct impacts on economic variables, highlighting the value of interpretable dimensions in understanding macroeconomic dynamics. The authors demonstrated that the way the model is constructed enables users to make connections between macroeconomic indicators and input words through the bias of types of words.

These interpretations establish a direct link between the financial world and the machine learning model through the importance of the identified word types. A financial analyst could use these explanations not only to verify the model’s outputs but also to gain deeper insights into the macroeconomic context. For instance, understanding how specific dimensions like “negative outlook” or “legal risk” drive certain predictions allows analysts to anticipate broader economic consequences

and make more informed decisions. A similar model using linear regression and text as input was proposed in Reference [127].

**5.1.2 Attention Mechanism.** Attention mechanisms in deep learning models, such as the conventional attention and the self-attention found in Transformers [106], offer interpretability by assessing the importance of input features [34, 44, 122, 126], internal features [22, 63, 129], or specific time periods [20, 44, 104]. Indeed, the attention mechanism is based on correctly weighting the inputs to make a better prediction. It provides an easy way to include the time importance in the interpretation.

In fact, the time importance is desirable for a financial analyst to understand which time frame the model utilizes for its predictions. The time-frame information helps him/her determine if there was a specific event that happened during that time. This event could be a significant decrease in a stock's price or a company sale. Such information not only helps the analyst figure out if the model's prediction could be incorrect but also provides a deeper understanding of the context. Attention models offer this kind of insight through their two-dimensional attention weights. Since attention models take sequential data with both a time dimension and a feature dimension as input, the attention weights also have these two dimensions. In fact, these weights indicate where the model is focusing its attention, on which features, and at which time points. The weights can then be extracted for model interpretation and are often visualized through heatmaps where one dimension represents time and the other represents features. This offers insights into the model's selective focus and its consideration of contextual relevance. To obtain the absolute time or feature importance, the weights can be summed according to the right dimension. We present some attention models that can be used to analyse feature and time importance of sequential input such as time series, words and events.

The **Interpretable-LSTM (ILSTM)** proposed in Reference [44] stands out for its ability to separate hidden states by features, enabling a clear understanding of their individual contributions to predictions. The model combines temporal and variable-wise attention mechanisms, which aggregate these hidden states for forecasting while providing insights into feature importance. In an experimental study, the model was used to predict the NASDAQ 100 index based on the performance of its constituent companies. The results highlighted the varying importance of each company across different time lags and epochs, with NTAP, FOXA, and TRIP emerging as the three most influential stocks. Interestingly, the feature importance rankings differ between the ILSTM and the benchmark model, raising questions about the alignment of model predictions with market realities and the potential reasons behind these discrepancies. Comparing the feature importance derived from these models with the official NASDAQ 100 weights could offer valuable insights into which model more accurately reflects market dynamics.

Similarly, a distinct adaptation of LSTM was presented in Reference [126]. The model is named **attention-based LSTM (AT-LSTM)** and was used to predict stock prices. The unique part of this model is its attention module placed before the LSTM. This module is used to weight the input features. The basic idea that the model relies on is that input features should not all have the same weight (as in a standard LSTM). Like a financial analyst would do, the model needs to weigh the information it receives to make a good prediction. In addition to improving the LSTM's performance, this module adds a layer of interpretability. Indeed, the user can visualize the importance of features from the attention weights.

The **Multi-modality graph neural network (MAGNN)** was presented in Reference [22]. MAGNN was used to predict financial time series. Unlike typical regression models, MAGNN uses not only time series as input but also events and news. These types of input are named "modalities" by the authors. The model mainly consists of attention mechanisms and graphs. There are

two types of attention in the model: intermodality attention and inner-modality attention. The intermodality attention is used to determine the importance of each modality depending on the different market situations, i.e., the different predicted returns. For example, the intermodality attention proved that all the modalities are important in the prediction. Also, it showed that the event modality seems to be more relevant in high-return situations, unlike the market and equity modalities, which are more important in low-return situations. This difference can be explained by the nature of news events, which often drive sudden and significant changes in equity prices. News stories explicitly highlight impactful events, such as corporate acquisitions, regulatory changes, or unexpected financial results, which can cause sharp price fluctuations. In contrast, during periods of lower returns, market and equity signals, which evolve more gradually, provide more stable and reliable information for prediction. The inner-modality attention, slightly less interpretable, is used to learn the relations between input sources and the target prediction.

For quarterly revenue prediction, the **Domain Adaptive Multi-Modality Neural Attention Network**, named Dandelion, was proposed in Reference [129] and was designed to be interpretable by end users while delivering good results in predicting company revenues. This model is unique due to three components. First, it exploits the interconnections between various modalities. The modalities were, in that paper, weather, finance, and news. Second, the model has the capability to “capture the high-level domain knowledge of stocks and understand the underlying importance and logic of modalities/variables across different domains” [129]. Finally, it explains outputs by revealing feature importance, showing relevant domain and time stamp details. The model’s trinity attention indicates the importance of features based on the task, modality, and time. For example, it indicated that the temporal importance increases over times. It suggests that recent information is more important than distant information, which makes sense. Also, it revealed that finance features are the most important among the weather, news, Wall Street, and web features.

While attention weights provide a glimpse into a model’s processing, it is not always a definitive representation of its reasoning, and its interpretability may vary across tasks. This will be discussed in the Section 5.4.

**5.1.3 Decision Trees.** Decision trees are among the most interpretable machine learning algorithms. A decision tree models decision making through a series of questions based on feature values, leading to a clear, treelike structure that can be easily visualized and understood. Each node in the tree represents a decision criterion, and each subsequent branch represents an outcome of that decision. The final leaf nodes provide the model’s prediction. Their transparent nature is reflected in that the reasoning behind any specific prediction can be traced back through the tree, making it straightforward to explain why a particular decision is made. This clarity in decision-making processes makes decision trees the candidates of choice in scenarios where interpretability is crucial. Usually, decision trees are interpretable by the rules they provide to understand the decision process (Section 5.2.1). However, decision trees can also be interpretable by providing the gain of each feature. The gain represents how much the node of a feature contributes to decreasing the uncertainty of the model during the training.

As an example of applying decision trees to financial time-series forecasting, Silva et al. [2021] used decision trees to predict the closing price of IBOVESPA with historical values as input and technical analysis. More specifically, they introduced the **Fuzzy Decision Tree-Fuzzy Time Series (FDT-FTS)** method for this task. As the name suggests, this approach consists of a fuzzy decision tree induced from the C4.5 algorithm [88]. The importance of the features is derived from the information gained during tree induction, highlighting RSI, the difference in moving averages, and the first difference series delayed by  $t - 1$ . Although these features do not have an explicit interpretation, they indicate that these technical indicators have a significant impact on the predictions.

**5.1.4 Graph.** Often, when graphs are incorporated within a model, they allow capturing relationships among features and estimating the importance of these features. This makes the model interpretable, as graphs are generally easy for humans to understand.

Using a textual network, Li et al. [2020] analyzed the relationship between a stock index and analysts' research reports. Although, it is not directly related to time-series prediction, their process predicts the stock movement using a sparse Laplacian logistic regression. This logistic regression is based on a graph that is constructed from analysts' research reports. Specifically, 56 most representative words, termed as "representative textual features," were extracted from the reports and used as the graph's nodes. An edge between two words was established if the two words were statistically frequent in the same window in a report. Subsequently, the graph was interpreted via the most influential words based on their centrality, the graph's density, closely interlinked subgroups, and so on. The importance of each word is presented in the article. "Imagine," "Pessimism," and "Effectiveness" are the words having the highest positive impact, while "Concern," "Design," and "Chengyu stock" have the highest negative impact. The negative words seem to have a higher impact than the positive words according to their importance coefficient.

Another interpretable model using graph was presented in Reference [28]. It is named **Knowledge-Driven Temporal Convolutional Network (KDTCN)**. The model takes stock price values and news as input and predicts stock movements.

The architecture is interpretable because of the effect of events on the prediction and the relationships between events that can be analyzed through graphs, like Reference [60]. Indeed, within their TCN framework, the effects of different events are easily extracted from the model, therefore significant events are identified in the graphs created in the KD module. The direct and indirect links between events can thus be analyzed, providing a deeper understanding of the decision-making context. For example, "[United Kingdom, votes to leave European Union]" has a high effect on the prediction of the stock to rise, according to Reference [28]. This demonstrates a clear link between global events and financial outcomes, providing users with valuable insights into how world events influence financial correlations.

Yet, to predict stock trends, Wang and Ye [2020] developed the **Graph-Based Interpretable Stock Trend Forecasting Framework (GIFF)**. GIFF is divided into three modules: the sequential embedding module, the graph residual module, and the forecasting module. A distinctive feature of this architecture lies in the graph residual module. This module is subdivided into three layers: the business-based layer, the industry-based layer, and the fully connected layer. In the first and second layers, connections are established between stocks that share the same businesses and industries, respectively, forming graphs. The last layer connects every pair of stocks. Since it is partitioned in this manner, the layers are interpretable and the results can be interpreted for each different industry. For instance, the authors compared the correlation between the predicted returns and the actual returns of the same industry (e.g., electronics). This allowed them to determine how well their model predicts for each economic sector and individual business. However, compared to Reference [60], this approach does not allow the direct measurement of the feature importance.

Similarly, the model proposed in Reference [50] does not explicitly compute the feature importance. Instead, it highlights relations between the inputs. This model, named **Multilevel Graph Attention Network (ML-GAT)**, predicts stock movements by taking as input numerical from Yahoo Finance website, textual from news, and relational data from Wikidata. Essentially, the model extracts features from various types of input: It encodes text data and price data and derives graphs from company relationships. Subsequently, the text and price data are incorporated into the graphs. Then, the price trend is forecast. This model is interpretable due to the insight that can be drawn from the created graph. The relationships reflect reality, since they are derived from real-world

data, allowing users to obtain a comprehensive visualization of the situation. Another recent example of interpretable model with graph is presented in Reference [109].

**5.1.5 Other Techniques.** Other techniques can provide feature importance through interpretable models. The ones reviewed here include fuzzy logic, masking methods, greedy algorithms, and hybrid techniques.

First, the model, named **Interpretable Intuitionistic Fuzzy Inference Model (IIFI)**, presented in Reference [108], combines intuitionistic fuzzy logic with deep learning for stock price prediction and stock trading prediction tasks. This fuzzy logic framework effectively captures feature importance. When predicting the most representative securities of the China A-Stock market, the IIFI identifies the most influential features as the sum of the ratio of the mean active buy turnover to the standard deviation of active turnover, the correlation between closing price and volume, and the fluctuation ratio of the buy order change ratio. Additionally, the model can differentiate the importance of features between rising stocks and falling stocks, acknowledging that the key drivers of stock behavior are not always the same in different market conditions. This observation aligns with established financial theories, reinforcing the idea that context-specific information drives market dynamics. The IIFI model confirms and quantifies this distinction, offering valuable insights into feature relevance across varying scenarios.

Second, the **Combined Time-View TabNet (CTV-TabNet)**, proposed in Reference [97], performs stock movement prediction using candlestick patterns and moving averages as inputs. A key feature of its architecture is the separation of short-term and long-term views possible according to a mask in the architecture, allowing their relative strengths to be extracted and analyzed. This design reveals whether the short-term or long-term components contribute more significantly to the prediction, offering insights into the temporal dynamics of stock price movements. It provides users with a better understanding of the duration of an event's impact on the stock market. Additionally, the architecture delivers valuable information on both feature importance and time importance, enhancing interpretability and enabling users to identify which factors and timeframes most influence stock predictions.

Third, to classify intraday stock movements, [Kumar and Haider \[2021\]](#) proposed an architecture that includes the greedy recursive feature elimination algorithm for calculating feature importance. This algorithm calculates the effect of eliminating subsets of features on accuracy to retain the subset of features that maximizes the accuracy. Thus, feature importance is extracted directly from the effect of the features on accuracy and can be analyzed.

Fourth, several models [124, 128] combine different techniques to provide feature importance. For example, the approach in Reference [124] leverages tree-based models (XGBoost, Random Forest, Decision Tree, etc.), a two-stage feature selection process, and a local interpretability module to compute global and local feature importance, as well as time importance. This enhances both the accuracy and interpretability of tree-based models. The model predicts the XOM price using internal features, which are technical indicators of XOM, and external features, which are time series unrelated to XOM. According to the selection method, the most important internal features were the weighted closing price and the balance of power. The transformed price features are all significant, because they rely on past prices, which influence future prices, as noted in References [124] and [80]. Based on Reference [11], the authors stated that the balance of power “implies the direction of a trend measuring the strength of buying against selling by assessing the ability of each side to drive prices to an extreme level” [124], which explains its high importance. The external feature analysis revealed that both SPE and VLO played key roles in prediction across all data segments, though local feature importance varied slightly. SPE was the most important feature in three of the four time segments. This study highlights that both internal and external features contribute

to predictive power, though internal features appear to be more effective. Another model, introduced in Reference [128], integrates regression techniques with financial theory to provide interpretation.

## 5.2 Decision Rules

Decision rules are another XAI principle that contributes to the interpretability of a model. They prescribe the conditions a model employs to make predictions, reflecting the patterns learned from the data. In some cases, these rules can be linked to financial theory [113]. By revealing these rules, users develop a clearer understanding of the model's operations and can find patterns that they might not otherwise guess. Decision rules are one of the most actionable and transparent XAI principles. They explicitly show the effect of an input (e.g., price, period, event, word) on the output. Financial analysts can use them to clearly understand why a decision was made by the model. These rules are also valuable for better understanding the financial context and can sometimes reveal or confirm fundamental financial theories, such as mean reversion. Because decision rules can be easily translated into plain language, they are straightforward to interpret for both financial data scientists and end-users, facilitating communication and alignment across teams.

There are two main techniques for providing decision rules: from a decision tree or from a rule-based module. Even though the decision tree is a type of model, it can be used within a complex model and retains its interpretable nature. However, a rule-based module is a component of a model or the entire model itself that contains the decision rules. It includes all specific methods for computing the rules such as rules derived from experts, learned rules, and a mixture of both. By convention, a rule-based module does not include decision trees.

**5.2.1 Decision Trees.** Decision trees is a technique that provides decision rules. Each decision tree's node represents a decision made by the model. The decision trees differ from other rule-based modules in the algorithm for trees construction. Such an algorithm often utilizes information gain from features to construct the trees. This gain can also be extracted to provide feature importance. According to Reference [16], decision trees must be relatively short to maintain a low number of rules and, consequently, be interpretable.

A way to use decision trees is to transform a complex model into a decision tree and then interpret the decision tree. Indeed, to predict stock price movements, the approach introduced in [113] uses a GRU and a GRU-Tree. The GRU-Tree is a GRU to which the regularized tree algorithm proposed in Reference [112] has been applied. This algorithm transforms a GRU into a decision tree. Thus, the GRU-Tree is intrinsically interpretable for the same reasons as a normal decision tree. The process is in three steps. First, the stock price is predicted with the GRU. Second, the regularized tree algorithm is applied and a decision tree, i.e., the GRU-Tree, is obtained from the GRU. Third, the GRU-Tree is interpreted. One interesting thing is that the authors found that some branches of the tree exposed the mean reversal, a known rule in finance. They also used the GRU-Tree to predict the stock price movements and computed the fidelity between the GRU and the GRU-Tree to be about 0.8. Therefore, GRU-Tree acts as an interpretable proxy model of GRU, allowing the predictions made by the GRU to be understood and analyzed in the form of a decision tree. To upgrade their model, the same group of authors improved their GRU-Tree with a L1-Orthogonal Regularization to form the L1-Orthogonal Regularized GRU Decision Tree [114]. This improvement increased the fidelity score to approximately 0.97,

Another, and simpler, way to use a decision tree is to incorporate it directly into the model. Li et al. [2021] proposed a complex architecture that predicts stock movements by taking as input prices, textual data from the Web, historical data on terrorist incidents, and socioeconomic data. Within the architecture, they attempted, from the given inputs, to determine if there has been

a significant incident that could impact the market, such as a terrorist attack, and subsequently predict stock movements. The model is interpretable, because the final decision is made based on a decision tree. Therefore, users can see the choices the model made to arrive at the prediction. For instance, the interpretations can be: “On 1/6/2009 and 1/7/2009, three news titles are detected to be about terrorist attacks, with key facts including countries (Israel), regions (Gaza), deaths (10, 42, 4), injuries (0, 0, 0), and weapons (firearms, others, others). They are cross-referenced with market and socioeconomic datasets and aggregated into a vector for 1/7/2009. The market is predicted to drop because: 1. S&P500 index drops today; 2. though the number of deaths is no more than 4, there are attacks in the last two days” [61]. This interpretation clearly illustrates the connection between the event and its effect on stock prices, offering users transparency into how specific features, such as the location, intensity, and frequency of incidents, are factored into the prediction.

Similarly, Carta et al. [2021b] proposed an interpretable approach for predicting the magnitude of stock price variations using a decision tree. The model integrates information from stock prices and financial news to identify patterns and relationships that drive stock movements. The approach relies on a lexicon-based analysis, where impactful words from financial news are identified and associated with stock price changes. These insights are then used to extract relevant features from the news, which are fed into a decision tree to predict whether stock price variations will be high or low. The interpretability of the model comes from its ability to reveal decision rules used in predictions and to highlight key sentences from the analyzed news that influenced the outcomes. For example, sentences containing impactful lexicon words linked to significant price movements can be presented, offering users a clear understanding of how textual information contributes to predictions.

**5.2.2 Rule-based.** Many complex models remain interpretable thanks to rules they contain. These rules often follow the Mamdani-type approach, which involves linguistic variables and fuzzy logic to represent knowledge in a human-readable format. Mamdani-type rules are particularly advantageous for interpretability as they provide understandable, transparent decision-making processes that can be easily translated into English and understood by humans. Some rules are initiated by experts [123] and others are learned from data with an algorithm [3, 13, 35, 48, 70, 89, 115, 116, 118].

An interesting rule-based model was introduced in Reference [89] that proposed an interpretable neuro-fuzzy approach using Mamdani-type rules for stock price prediction. The model employs clustering techniques to identify key patterns in the data, which are then transformed into interpretable rules. To maintain transparency, the model applies specific training constraints during backpropagation and includes a rule selection process to reduce complexity while preserving accuracy. The results achieved by the model are comparable to ANFIS [53], demonstrating its ability to balance predictive accuracy and interpretability [89]. This balance makes it a valuable tool for financial professionals seeking both reliable predictions and clear explanations.

Another way to make an interpretable model based on rules is to integrate rules into an already-known system. For instance, fuzzy logic was added to a Hammerstein–Wiener system to form the **Neural Fuzzy Hammerstein–Wiener (NFHW)** network [115]. It creates Mamdani-type fuzzy decision rules, making it more transparent than a conventional Hammerstein–Wiener. The most frequently used rules are also identified. The model was applied to stock prediction and the results were compared with other regression models based on fuzzy logic. The NFHW outperformed all these models. In another paper [116] by the same authors, an interpretable embedded convolutional fuzzy neural network was made with an architecture similar to the NFHW. On financial data, the model achieved better performance in predicting the S&P 500 index compared to other baseline models presented in the study. An example of a Mamdani-type rule given by the model is

“If the price on Day 1 is normal, that of Day 2 is low, that of Day 3 is low, that of Day 4 is low, and that of the Current Day is normal, then the closing price of the next trading day is predicted as low” [116]. These linguistic rules allow the user to directly understand the decision process behind the predictions. For example, a portfolio manager could use these insights to better understand how specific sequences of price movements influence future trends. This understanding would enable them to adjust portfolio allocations proactively, mitigating potential risks or capitalizing on anticipated market movements.

Instead of relying on a complex architecture, Ferdaus et al. [2022] proposed a new algorithm for learning rules, namely the **Type-2 Parsimonious Learning Machine (T2-PALM)**, an interpretable rule-based model designed for predicting stock index fluctuations. To limit the number of rules and maintain clarity, the model uses evolutionary algorithms to refine and simplify its rule set. This ensures the model remains interpretable without compromising predictive accuracy.

In contrast to other models, the rules of the **Hierarchical Belief Rule Base (HBRB-I)** [123] originate from experts. HBRB-I consists of multiple **Belief Rule Base (BRB)** modules, a rule-based decision-making system used to predict stock movements. To enhance interpretability, 11 specific criteria related to modeling, inference, and optimization are applied. These criteria emphasize that the system should be simple, consistent, and have clear semantics. They also suggest that the inference process should be transparent and that expert knowledge should be reasonably incorporated [123]. Within HBRB, the BRB rules are initially defined with expert input. The BRB-I module then learns to adjust these rules by optimizing their weights. These weights can also be analyzed based on their importance. Another interpretable model that provides decision rules is presented in Reference [13].

### 5.3 Trends

Model interpretability has also been investigated from the perspective of trend analysis. Indeed, the trend is an important component of a time series that can be decomposed and analyzed in depth to extract significant information. It helps the financial analysts to better understand the underlying patterns in the time series, identify long-term market directions, and detect shifts or anomalies that could influence financial outcomes. The models presented in this section analyze time series to identify trends in the data, characterize them, and then link them to the prediction.

A GRU method based on a HPFilter was introduced in Reference [119] to predict stock prices. The HPFilter is an approach employed in macroeconomics for trend analysis. In the architecture, it allows for the decomposition of opening prices, closing prices, minimum prices and maximum prices into two components, i.e., long-term trend and short-term volatility. Each of these components is modelled by a specific GRU: SGRU for short-term and LGRU for long-term. The output of the components are combined to form the final prediction. This architecture merges the concept that a time series is composed of trends and volatility with the high performance of GRUs, achieving both accuracy and interpretability. Indeed, the model is more transparent than deep neural networks due to the trend decomposition, which allows the short-term fluctuations and the long-term trend to be visualized. Also, the combination of two models, with one GRU predicting the trend and another predicting the volatility, surpasses the accuracy of a single GRU [119].

Similarly, the N-BEATS model from Reference [79] enables the interpretation of trends and seasonality. In general, N-BEATS, built for univariate time-series prediction, is not interpretable due to its complex architecture. However, the authors introduced a special setup with constraints aligned to trends and seasonality, making the model interpretable without reducing performance. This setup allows trend and seasonality decomposition. Trends and seasonality were extracted separately from the model’s predictions and analyzed in charts alongside the forecasts.

Just as trends compose time series, both linear and nonlinear relationships also play a key role in their structure. This concept, known as “hybrid prediction” [125], suggests that a time series can be seen as a combination of a linear component and a nonlinear component. These two parts can be isolated and analyzed separately, offering valuable insights into the prediction process. In this approach, the linear part is modelled with ARIMA, while the nonlinear part is modelled with a neural network. Residuals are defined as the difference between the actual prediction and the linear part. Panja et al. [2022] applied this principle in their **Probabilistic Autoregressive Neural Network Framework (PARNN)** for time-series prediction. The interpretability of this model comes from the transparent division of responsibilities between ARIMA and the neural network. ARIMA offers clear insights into autoregression, moving averages, and integration, making its predictions understandable. While the overall framework might not be considered highly interpretable, the inclusion of ARIMA introduces a layer of transparency. Another hybrid model is presented in Reference [49].

#### 5.4 Interpretability Evaluation

How to evaluate interpretability is an open issue in the machine learning community. Existing approaches to evaluating the interpretability deal mainly with one of the two following aspects: the interpretability of a model, i.e., how much the model is interpretable, and the reliability of the interpretations, i.e., how closely the XAI principles reflect the underlying logic of the model.

Ideally, we would like to have some metrics that allow to directly measure the level of interpretability of a model. However, according to our knowledge, there is no such metrics function capable of doing that. Instead, existing work such as that in Reference [64] suggests three interpretability dimensions to be considered. The first one is the simulatability. This concept refers to the ability of a model to be entirely inspected by a human at once. It is the simplicity of a model. For example, a linear model with low-dimensional features is more interpretable than a linear model with high-dimensional features according to simulatability. This concept can be a bit subjective, because everyone does not have the same level of analysis capacity. Indeed, if we want to measure the simulatability, then we need to quantify the properties (number of parameters, number of sub-modules, number of layers, number of different sub-model, etc.) of the model. Almost all the models presented in this survey would have a low score in simulatability due to their complexity and their high number of modules. The second dimension proposed in Reference [64] is the decomposability. It is the ability of the parts (input, parameters and calculations) of a model having an intuitive explanation. For example, the parameters of the linear models represent the strength (feature importance) of the association between the features and the prediction. Another example is the nodes of decision trees that represent interpretable rules. The decomposability requires the input to be intuitively interpretable. Indeed, it would be hard to analyze the importance of a feature that is noninterpretable itself. The interpretable models presented in this article would have different scores of decomposability, because the percentage of interpretable parts is different among different models. The third dimension of interpretability according to Reference [64] is the algorithm transparency. This notion is the ability of the learning algorithm of a model to be simple, understandable by humans. Also, it includes the ability of a model to have a unique solution. As an example, linear models have this ability, unlike deep neural networks.

Evaluation of the reliability of the information provided by interpretable models is also a challenge. Since the models employed in practice are usually complex, the interpretations from such models may not be reliable. In fact, as we discussed earlier, in a complex interpretable model, the XAI principles result from the use of some technique that is placed in some part of the model’s decision process. When this part is directly linked to prediction, such as at the end of the decision process as presented in Reference [74], it allows illustrating how the prediction is made. For

example, if the prediction is computed by a linear sum that includes interpretable features, either internal or from input, then the weights of the linear sum directly represent the importance of these features. While the process to obtain the feature weights may be incomprehensible (opaque), the relationship between the features and the prediction can be understood without ambiguity. As long as the features mean something for the user, these weights appear to be reliable.

However, if the use of the technique is followed by several other operations, then the reliability of the information generated from the technique can be seriously affected. Indeed, sometimes the component that measures feature importance is at the beginning of the model's process, for example, the attention mechanism as in Reference [126]. This component allows the model to weight the input before processing it and makes the model interpretable. As the authors in [20, 22, 44, 63, 104, 122, 126, 129] suggested, these weights could be extracted and analyzed as the feature importance. Nevertheless, since processing steps are operated between this interpretable component and the final prediction, it is questioned about whether the feature importance will remain consistent after all these operations. It is possible that the model might learn to ignore those weights and instead learn the patterns of the data in another way. Consequently, the feature importance might have changed after these operations. This means that even if there's a module responsible for weighting the input at the beginning of the model, subsequent operations can change this weighting when it comes to prediction. Therefore, even though neural networks having an attention mechanism at the beginning like Zhang et al. [2019] are considered interpretable, the reliability of the interpretation should be questioned.

Going somewhat beyond the time-series models, several interesting works have been done by the NLP community on how to determine if the attention weights truly represent the feature importance. Their conclusions are not affirmative. Indeed, the attention weights are not considered as the best option for ranking features according to Reference [98]. The authors of this article show that the attention weights contain some useful information about the feature importance, but the information may not be enough to correctly rank the features. Also, the capacity of the attention weights is also questioned in Reference [52], where the attention weights do not confirm the feature importance. Indeed, feature importance from attention weights shows a low correlation with the one obtained from other methods, such as gradient-based and leave-one-out methods. Even if benchmark methods do not represent the truth, then the experience shows that the reliability of the attention weights is questionable. If we extend the logic of the definition of the "explanation" in Reference [111], then the attention weights could be reliable, depending on the definition of the reliability. If a reliable interpretation has to be plausible and faithful, then attention weights would not be reliable. However, if it only has to be plausible, then they would be one. These divergent results suggest that the reliability of the attention weights depends on the context, the definition and the testing method. It could thus be different in a task of forecasting time series. These analyses show that the reliability of the interpretation of complex models must be challenged, and their use must be done with caution.

In general, there is no standard method to measure the reliability of the interpretations. However, at this point, the interpretations of an interpretable model and the explanations of an explainable model are essentially equivalent, because the difference between an interpretation and an explanation results from their source, not from their nature. Some ideas and approaches to evaluate explanations are presented in Section 6.3. They could be applied to interpretations.

## 6 Explainable Models and Explainability Methods

Explaining a model using an explainability method is a common approach. Once a model has been selected or validated for its performance, it is often more practical to explain it using an external method rather than redesigning it to make it inherently interpretable. Additionally, many

explainability methods are readily available in popular libraries, making them accessible and easy to implement. However, unlike interpretable models, the explanations generated by an explainability method are not intrinsically tied to the model itself. As a result, their fidelity can be questioned, as will be discussed at the end of this section.

This section presents explainability methods and their application contexts, emphasizing their role over the models themselves. Since this article focuses on XAI, much emphasis has been placed on the explainability methods rather than on the models themselves. Some financial insights that the authors derived from these methods are also presented. The explainability methods are grouped according to their XAI principles, such as feature importance and visual explanation, and according to the technique that they use to extract explanations from models. Finally, evaluation of explainability is discussed.

## 6.1 Feature and Time Importance

With a similar objective as interpretable models, explainable models employ a technique to elucidate the impact of features on the output. Basically, two techniques are used to measure the feature importance of models: the propagation technique and the perturbation technique. Methods using the propagation technique track features through the model to gather information about their influence and observe the outcomes, while methods using the perturbation technique alter the inputs and compute the difference in results to quantify the effect of the features.

**6.1.1 Propagation.** The propagation technique, designed primarily for analyzing neural network based models, makes use of the information provided by the gradient or the neurons. Indeed, neural networks fundamentally rely on a sequence of weighted multiplications. These multiplications can be followed through the model from the input layer to the output layer during the forward pass. They can also be traced from the output to the input layer during the backward pass. The composition of multiplication allows neural networks to be differentiable. The propagation technique is based on these properties, i.e., the multiplicative composition and the differentiability of neural networks. Similarly to the propagation of information in the forward and backward passes of neural networks, the XAI methods using propagation can compute a score for each neuron from the output layer to the features layer to determine the importance of the features. An example of such a propagation technique is **Layer-Wise Relevance Propagation (LRP)** proposed by Bach et al. [4]. Moreover, these XAI methods can also take advantage of these passes to compute feature importance such as **Integrated Gradient (IG)** [103] thanks to the differentiability of neural networks.

A neural network for predicting stock returns, named Deep Factor network, was proposed in [75] that employed LRP [4] as the explainability method. LRP propagates relevance scores, representing the feature importance, from the output layer of the network back to the input layer like the backpropagation. The relevance scores of neurons of each layer are computed from the weights of the layer and the relevance scores of the previous layer [4, 75]. The final result of this propagation is the feature relevance/importance of financial factors including risk, quality, momentum, value, and size as the input features were grouped by financial themes (factors). The feature importance given by the LRP has been compared with correlation coefficients including Kendall and Spearman. Their rankings did not align. For example, LRP indicated a high contribution from the value factor, while the correlation coefficients suggested low impact. This verification does not affirm or dismiss the reliability of LRP, but it highlights a divergence between the two methods. As an explainability method, LRP will need more rigorous testing in a financial context. The Deep Factor network has also been compared to other models, including linear ones and proved to be more accurate. Finally, Reference [75] includes also a method to determine the importance of the financial factors through different time periods.

As mentioned earlier, integrated gradient [103] is another explanation technique that uses the differentiability of the neural networks to explain them. Indeed, in its process, the partial derivatives of the model are computed. The method consists in computing the integral of the partial derivative along the line between the input (feature) point  $x$  and a baseline point  $x'$ , which, for time-series prediction, is defined as the average of  $n$  most recent data points. The integrated gradient of a specific feature can be interpreted as the component that contributes to adjusting the model's prediction from  $x'$  to  $x$ . It has been applied to explain the hierarchical neural network in Reference [23]. This hierarchical neural network takes both textual data and numerical data as input to predict the SET (Thailand stock market) index, which causes difficulties to LRP based method for model explanation. The integrated gradient allows to extract the input words that positively and negatively impacted the stock prediction.

**6.1.2 Perturbation Technique.** The perturbation is a well-known technique for obtaining the feature importance of a black box model. The concept behind this technique is straightforward. First, a prediction is made using all features of the trained model for a given input point. Then, within the set of all features, the feature of interest is perturbed, and another prediction is generated using this perturbed feature along with all other features for the same input point. A comparison, basically in terms of their differences, between these two predictions is made to estimate the importance of the features. The methods that employ the perturbation technique can be classified into classic methods and sophisticated methods. The classic methods are the ones that apply directly this concept and the sophisticated methods rely on the perturbation technique. The sophisticated methods include SHAP [38] and LIME [91].

In general, the methods using perturbation are easy to apply. Indeed, they are model agnostic, i.e., they do not depend on the model, so they can be applied on all types of models. The classic methods are easy to code, while the sophisticated ones, like SHAP or LIME, have their library already implemented in Python, which makes them readily applicable. The output of these methods is the feature importance that can help understand the relation between the features and the output of a model. It can also be used to make feature selection according to Reference [15]. As an example, the popular and highly effective model for predicting financial time series, XGboost, has been explained by SHAP [29, 39, 110]. The perturbation technique can also be used to compute a confidence level of a prediction [19].

**Classic Perturbation Methods.** As we mentioned earlier, the idea behind the classic perturbation methods is to modify a feature, make a prediction, and then compare the result with the prediction using the original feature through a dissimilarity measure. This process can be applied to a single prediction or a set of predictions. There are several ways to modify the feature. The first approach involves randomly shuffling the data of the feature with historical feature values, known as Permutation Importance, as proposed in Reference [12]. The second method is to add noise to the feature values, as discussed in Reference [15]. Determining the appropriate magnitude of the noise can be challenging for this method. The third method is to zero out the feature, referred to as “ablation” [38]. In the context of time series, setting a feature to zero may not be meaningful. The common solution to this issue is to replace the value with the average value of the feature over the last  $n$  days.

Classic perturbation methods, such as ablation, added noise, and permutation importance, were applied on recurrent networks (RNN, LSTM, and GRU) in Reference [38] and in Reference [17]. We discuss here the work reported in Reference [38]. The models take, as input, high, open, low close, volume and adjusted close to predict stock prices. The integrated gradient method was also tested. The baseline used for IG was the average of the last 91 days, on the gradients. Based on the findings, the authors concluded that these techniques are applicable to recurrent networks employed in

financial time-series prediction to provide explanation of their complex architecture. Also, they found that the methods are complimentary. It makes sense, because each method was applied in a way to extract some different information. Indeed, the ablation and IG aim to extract the time and feature importance. More specifically, it was possible to determine with IG the sign of the correlation between the feature at a specific time with the output, but not with the ablation method. The added noise was used to extract only the time importance, and the permutation importance only extracted the feature importance. The results of all the four methods show that the importance of the volume is low as compared to the other features. The importance of the open, high, low, close and adjusted closed features are mostly equal for all models and methods. This is explained by the fact that the volume does not refer to the price but to the number of stocks traded. Indeed, if a stock is bought or sold, then the volume value will increase, giving no information about the price of the stock. Also, as expected, the most recent days closest to the prediction are more important than the more distant ones according to ablation, integrated gradients and added noise. The authors have presented a complete analysis of these methods through the different models. The importances of ablation and permutation importance remain relatively constant across the models, while the effects of IG and added noise vary significantly.

*SHAP.* SHAP is a sophisticated method for explaining the output of machine learning models and is based on the perturbation [38]. It is rooted in game theory, where it assigns each feature an importance value for a particular prediction. SHAP evaluates the effect of a feature by considering its absence in all possible combinations of features and calculates the difference in predictions compared to when the feature is present. The values computed by SHAP, named SHAP values, measure the average contribution of a feature to every prediction, ensuring consistent and fairly distributed feature importance scores. These values can be computed for any model and provide intuitive insights into its behavior. By decomposing a prediction into the sum of SHAP values for each feature, one can understand how differently features contribute, either positively or negatively, to the prediction. It was proved that SHAP is the only additive model that respects the properties of local accuracy, missingness and consistency [65].

There are several implemented versions of SHAP depending on the model to which it is applied. For example, Kernel SHAP can be applied on all types of models. However, the independence between features is assumed to compute Kernel SHAP. Another example is TreeSHAP [65], which is the SHAP method adapted for tree-based models. As the tree gradient boosting model XGBoost is popular in financial time-series forecasting, TreeSHAP is useful to compute its feature importance. While the global importance of a feature can be derived from XGBoost's gain metric, it does not provide feature importance for individual predictions, as SHAP does. Indeed, Weng et al. [2022] analyzed the role of financial pressure in predicting the volatility of healthcare stocks with tree-based models, including XGBoost, and TreeSHAP. The models take as input past volatility of health care stocks from China and USA and the financial pressure measured by **Office of Financial Research Financial Pressure Indexes (OFR FSI)**. The OFR FSI contains five financial categories such as credit, equity valuations, safe assets, funding deviation, and volatility. The target value was the future volatility of health care stock of both countries. TreeSHAP was used to calculate the relative feature effect, that is, the SHAP values divided by a benchmark. The relative feature effect was utilized in four analyses. Even if the global importance of the feature can be obtained by the gain of XGBoost, it does not give the importance for each prediction, as SHAP does it.

First, the effect of financial pressure (OFR FSI) on the volatility of health care stocks was analyzed for China and USA. The health care stocks of China seemed to be more sensitive to financial pressure than the health care stocks of the USA due to their greater contribution to the prediction. Based on the theory in References [37, 68], the authors explained this result by the

fact that “emerging countries pursue quicker and aggressive contingent response strategies in response to external financial pressures” [110]. Second, the importance of the five categories of financial pressure on the volatility of health care stocks was computed for the USA and China. The equity indicator was found the most important indicator for both countries, while the safe asset indicator is the least important one. The authors explained that by the fact that “the extent of investor confidence and risk appetite directly results in the extent of market illiquidity and increased asymmetric information” [110], according to Reference [47], and also that the higher the cash flow, the more stable the stock market will be. Third, the different period lags of the OFR FSI were analyzed. The seven-period lag was the most important for both countries, but the second most important was six-period lag for the China and one-period lag for the USA. It shows that countries respond at different speeds to financial pressure [110]. Fourth, the contribution of the seven-period lag of the five financial pressure indicators was calculated. It shows that the distribution of the contribution of different period lags changes through indicators. Indeed, the effect of the different period lags is similar to the credit, while it is very heterogeneous for the equity indicators for both countries. However, the period lag for each indicator is different for the USA compared to China. Indeed, they found that “the volatility forecasting of Chinese health care stocks is mainly influenced by each indicator of financial pressure at the late and medium period lags; however, it was influenced at the front and medium period lags for the USA” [110].

These analyses are concrete examples of how we can use XAI to understand financial patterns. Originally, SHAP computes local feature importance. Then, global feature importance can be computed by doing the average of the SHAP values over the training sample of a model like that in Reference [117] did to understand which features were important in the training of their model and to do a feature selection after. Other examples of use of SHAP to explain a model for predicting financial time series are presented in References [29, 30, 39, 41–43, 46, 51, 73, 76, 78, 121].

*LIME.* LIME is a method in Reference [91] to explain the predictions of any black-box machine learning model. It works by approximating the complex model locally around the prediction of interest using an interpretable model, usually a linear model. It assumes that the surrogate and black-box models are equal at the point of interest. By perturbing the input data, generating new samples, fitting a surrogate model, and interpreting it, the black-box model can be explained. This provides a clear explanation, showing how each feature affects the model’s decision for a given instance.

Contrary to SHAP, LIME cannot be used to explain a model globally. This is one of the reasons why Ghosh et al. [2023b] used LIME to obtain local feature importance and SHAP to offer the global feature importance. Their goal was to predict Spanish and Indian stock during the COVID-19 pandemic and to understand the effects of macroeconomic indicators and media chatter on these predictions. The media chatter-linked is measured by some indices such as Panic Index and Media Hype Index. Many financial conclusions were made by the authors. Indeed, they found that the Spanish stocks seem to be less predictable than the Indians stocks. There would be a higher inefficiency in the Indian market than in the Spanish one. Also, they found that the global and local macroeconomic indicators were important in the prediction. The authors suggested that the media chatter should also be considered to “decode the temporal evolutionary pattern of the underlying assets” [40]. Even though the pandemic brought about significant disruptions and uncertainties, it did not fundamentally alter the underlying inefficiencies that exist in financial markets due to the importance of macroeconomics reflectors, but the media chatter emerges as an important feature in the prediction of the stocks of the two countries. Indeed, media hype and sentiment should be considered when making short-term investments in the equity market [40]. This work highlights the benefits of using LIME to understand financial contexts. Another example of LIME with time-series forecasting is Reference [90].

An issue with LIME is that it is hard to define the region around the prediction, especially when it is a time series. That is why Reference [96] introduced a new technique, named TS-MULE, for segmenting and replacing the time series. Another version of LIME was presented in Reference [19] to apply it in an unconventional manner. Indeed, it is used not to show feature importance but to transform the binary output of the random forest into an “explanation prediction probability.” This probability allows us to assign a confidence level to the base classification. It informs the model and the user whether to rely on the prediction based on a chosen confidence threshold. As the confidence level is used to make final predictions, the architecture could be classified as interpretable. However, it would make more sense to put it in the LIME section with the other variants of the LIME method. Furthermore, another special case of LIME is when a decision tree is introduced in the surrogate model. In this case, in addition to generating feature importance, this model has the ability to produce local decision rules [58, 87]. For instance, in Reference [87], the rules were extracted and tested according to their investment utility. The best rules were kept according to this test.

## 6.2 Visual Explanations

Visualizing the inputs, core, or outputs of a model is an effective way to understand its predictions and decision-making process. This principle, named as visual explanations, encompasses all visual information that helps users understand a model’s behavior. In general, this principle is inherently understandable to humans due to our natural affinity for visual representations. Compared to other XAI principles that emphasize what information is presented, this one focuses on how the information is presented to ensure clarity and human understandability. It is often accompanied by other principles, such as feature importance, which highlights the contribution of individual features to the model’s predictions.

One widely used technique for delivering visual explanations is the dashboard. A dashboard serves as an interactive interface that consolidates diverse types of information in a coherent and user-friendly manner. It can display historical stock trends, technical indicators, real-time stock movement updates, and relevant news articles associated with prediction anomalies or spikes. Predictions can also be plotted against industry benchmarks, such as the S&P 500, to assess relative performance. Dashboards often allow users to customize visualizations, adjust model parameters, and explore predictions across different time horizons. By presenting this information in an organized and accessible format, dashboards provide a comprehensive view of both the model’s behavior and the broader financial context.

Another used visualization technique is the feature importance plot, which displays how much each variable contributes to a model’s predictions. These plots allow financial analysts to identify the key drivers behind a prediction and understand which variables, such as interest rates, trading volume, or economic indicators, had the most significant impact on the outcome. Although dashboards often incorporate feature importance plots as part of their visual components, this tool remains standalone techniques that can be used independently to provide focused insights.

Two dashboards are presented in this section. First, DeepVIX [25] is an interactive dashboard designed to explain LSTM networks. The interface shows five different information: training parameters, comparison between training and test loss, configuration parameters, LSTM architecture, and interactive features. In fact, the interactive features part allows users to view the four types of weights (input gate, forget gate, cell state, and output gate) throughout the network and iterations. These weights are represented visually via plots corresponding to the structure of the network, offering an intuitive way to track their evolution. Its primary aim is to offer a deeper understanding of the LSTM, enabling better optimization. With this interface, it is easy to observe changes in the parameters throughout the iterations and relate these changes to machine learning theory.

Although this dashboard does not provide direct financial insights, it significantly enhances the explainability of the model, aligning well with the definition of explainability.

Another popular interactive interface is DeepClue, as introduced in Reference [100]. It is a dashboard designed to elucidate the hierarchical neural network also proposed in that paper. This network takes financial news headlines as input and predicts stock prices. It contains many tools that a financial professional could use to facilitate their work. A portfolio manager, for example, could use this dashboard to make better trading decisions by analyzing the relevance scores, computed by LRP [4], of key factors affecting stock movements. By visualizing these scores in feature importance plots, he can identify which textual variables are driving the model's predictions. The stock timelines and document lists displayed on the dashboard provide a chronological view of significant events and market reactions, helping the manager understand how certain news or market conditions correlate with stock performance. Additionally, the list of documents related to selected factors offers a detailed contextual analysis, enabling the manager to verify the rationale behind specific predictions. For instance, if a stock's prediction indicates an imminent rise due to positive sentiment in financial news, then the trader can quickly cross-reference these insights with the source documents directly displayed on the dashboard. Since the inputs are primarily interpretable text data, the explanations derived from the dashboard are inherently clearer. This clarity enables the manager to integrate model insights with their financial expertise, supporting informed and contextually appropriate decisions.

In practice, DeepClue was used to upgrade the model and to verify a prediction. Indeed, with the help of a trader and a stock analyst, the model's decisions were evaluated in Reference [100]. The trader analyzed the best moments to make buy/sell a specific stock, namely during significant increases or decreases. He analyzed the important words and identified a key cluster of words that made sense for him and for the prediction. The interface showed him the news from the cluster's source, thus enabling verification of the prediction's validity. The analyst, however, found a deficiency in the model with the help of DeepClue: "the afterhour trading is not considered in the training phase" [100]. Furthermore, Deepclue also helped the analyst discover that the model was overfitting in a case where the inputs were social media words. It is a true example where XAI increases the user's confidence in the model by linking the prediction to financial theory. Other dashboards are Hawkeye from Reference [14] and the ones presented in References [5, 54].

### 6.3 Explainability Evaluation

Similarly to interpretability evaluation, there is no established standard for explainability evaluation. As the concept of evaluation is large, it must be defined. Also, the evaluation could depend on the XAI principles of the methods. In this subsection, we present a few main criteria and approaches in this area.

Belle and Papantonis [2023b] and Robnik-Šikonja and Bohanec [2018] highlight, among other things, the following evaluation criteria for explainability: comprehensibility, fidelity, accuracy, stability, and completeness. Comprehensibility refers to the readability and the size of the explanation. Readability may depend on the user and the context (e.g., financial analyst vs. general public), while the size of the explanation relates to aspects such as the number of features or the number of rules. Fidelity describes how closely the explanations mirror the behavior of the prediction model. Accuracy refers to the ability of an explanation to generalize to unseen instances. Stability represents the consistency of an explanation when applied to similar instances. Some of these criteria were originally defined in the context of rule extraction in Reference [99].

They are theoretical concepts that make sense but are difficult to quantify. Another way to evaluate the explainability is to inspect the effect of the explainable method on a real task. Indeed,

a study in Reference [18] on the trust in XAI during high-uncertainty decisions was made. The authors examined the effects of AI explanations on user trust when the AI displays high versus low confidence levels. The study compared three types of explanations: inductive (based on local examples), abductive (highlighting local feature importance), and deductive (local rule-based explanations). Their findings suggest that explanations diminish performance when the model projects low confidence. Conversely, in scenarios where the model is highly confident, both abductive and deductive explanations enhance the user task performance compared to inductive explanations.

Comparative analysis [105] is also a way to evaluate explainability. It highlights the divergence or convergence of the methods. However, with this type of analysis, it is not possible to determine which method is better.

Two interesting measures for evaluating explainability techniques that show the importance of features in a time-series prediction context are introduced in Reference [81]. They are named **Area Over the Perturbation Curve for Regression (AOPCR)** and **Ablation Percentage Threshold (APT)**. The AOPCR measures the effect of removing a certain number of features, while the APT measures the percentage of features that need to be removed before crossing a certain threshold. Using these measures, the authors compared SHAP with intuitive methods, a global average replacement explainability method, and a random method. Specifically, the values of importance of features calculated by these methods were compared. The analysis was performed on different models. They concluded that, in general, SHAP outperformed the other methods according to their measure. Completeness is another criterion introduced in Reference [27], describing the property of an explanation model to be equivalent to the prediction model. The authors also proposed a metric to measure it.

## 7 Further Characterization of XAI Models

This section consolidates all the surveyed models in Table 1, including those not explicitly described in the main text, either because they did not fully meet the methodological criteria or due to space limitations. Approaches are classified according to seven criteria following the column of the Table 1. The *year* column indicates the year of publication. The *type* column specifies whether the approach is an explainable (E) method or an interpretable (I) model. The XAI principles—feature importance (FI), decision rules (DR), visual explanation (VE), confidence level (CL), and trends (T)—refer to the type of explainable information provided, as defined earlier in this article. The *scope* [6, 26] of the explanation or interpretation denotes its focus, which can be either global (GB) or local (LO). A *local explanation* provides insights into a single prediction, while a *global explanation* offers an understanding of the model's behavior across the entire dataset. The *technique* column describes the interpretable or explainable method employed to deliver the XAI principles, as detailed in Section 4. The *model dependency* [6, 26] criterion indicates whether the approach depends on a specific model architecture or can be applied universally. An approach is considered model specific (MS) if it relies on a particular model structure, while it is classified as model-agnostic (MA) if it can be applied across different types of models. Naturally, *interpretable models* are inherently *model specific*. Finally, the *financial applications* column illustrates the contexts in which these approaches have been applied. If no specific source is cited, then it indicates that the application was demonstrated directly in the paper introducing the approach. Otherwise, a reference is provided to highlight an external financial application of the approach. As *explainability methods* (and some *interpretable models*) can often be applied in diverse contexts, this criterion serves as an *example of application context* rather than a strict classification tied to the approach itself. Additionally, Table 2 provides all the abbreviations used in Table 1 for clarity.

Table 1. Characterization Table

| XAI Approach [Pub.]       | Year | Type | XAI Prin. | Scope  | Meth.    | Model Dep. | Financial Applications                       |
|---------------------------|------|------|-----------|--------|----------|------------|----------------------------------------------|
| LIME [91]                 | 2016 | E    | FI        | LO     | PER, APP | MA         | Stock Price Prediction [40, 42, 90]          |
| SHAP [65, 66]             | 2017 | E    | FI        | LO, GB | PER      | MA         | [29, 30, 39, 41–43, 46, 51, 73, 76, 78, 121] |
| LIME (nonusual) [19]      | 2023 | E    | CL        | LO     | PER      | MS         | Stock Mov. Prediction                        |
| DEMUX [33]                | 2022 | E    | FI        | LO     | PER      | MA         |                                              |
| Dynamask [24]             | 2021 | E    | FI        | LO     | PER      | MA         |                                              |
| DeepVix [25]              | 2020 | E    | FI        | GB     | VE       | MS         | Stock Price Prediction                       |
| LRP [4]                   | 2015 | E    | FI        | LO     | PRO      | MS         | Stock Price Prediction [75]                  |
| DeepClue [100]            | 2019 | E    | VE        | GB     | VE       | MS         | Stock Price Prediction                       |
| Integrated Gradient [103] | 2017 | E    | FI        | GB     | PRO      | MS         | Stock Price Prediction [23, 38]              |
| Permutation Imp. [1]      | 2010 | E    | FI        | LO, GB | PER      | MA         | Stock Price Prediction [38]                  |
| Ablation, Noising         |      | E    | FI        | LO, GB | PER      | MA         | Stock Price Prediction [38]                  |
| LMTE [58]                 | 2020 | E    | FI, DR    | LO     | PER      | MA         | Loan Lending Decision                        |
| LSTM-Forest [86]          | 2022 | E    | FI        | GB     | PER      | MS         | Stock Mov./Price Prediction [2]              |
| ALE [2]                   | 2019 | E    | FI        | LO, GB | PER      | MA         |                                              |
| ALE range [62]            | 2022 | E    | FI        | LO, GB | PER      | MA         | Options Price prediction [62]                |
| Hawkeye [14]              | 2021 | E    | VE        | LO, GB | OTH      | MA         | Stock Mov. Prediction, Stock Trading [14]    |
| [5]                       | 2021 | E    | VE        | LO, GB | OTH      | MA         | Stock Mov. Prediction [5]                    |
| GRU-TREE [112, 113]       | 2017 | I    | DR        | LO, GB | DT       | MS         | Stock Mov. Prediction                        |
| [74]                      | 2022 | I    | FI        | LO     | LR       | MS         | Stock Price Prediction                       |
| [63]                      | 2022 | I    | FI        | LO, GB | ATT      | MS         | Stock Price Prediction                       |
| CF [48]                   | 2020 | I    | DR        | LO, GB | FL, DR   | MS         | Stock Mov. Prediction                        |
| [116]                     | 2022 | I    | DR        | LO, GB | FL, DR   | MS         | Stock Price Prediction                       |
| HBRB-I [123]              | 2023 | I    | DR        | LO, GB | DR, FL   | MS         | Stock Mov. Prediction                        |
| NFW [115]                 | 2021 | I    | DR        | LO, GB | DR, FL   | MS         | Stock Price Prediction                       |
| [89]                      | 2019 | I    | DR        | LO, GB | DR, FL   | MS         | Stock Price Prediction                       |
| DCAFS [70]                | 2021 | I    | DR        | LO, GB | DR, FL   | MS         | Stock Price Prediction                       |
| [118]                     | 2019 | I    | DR        | LO, GB | OTH      | MS         | Portfolio Allocation Prediction              |
| [16]                      | 2021 | I    | DR        | LO, GB | DT       | MS         | Stock Mov. Prediction                        |
| N-BEATS [79]              | 2020 | I    | T         | LO     | OTH      | MS         | Stock Price Prediction                       |
| PARNN [84]                | 2022 | I    | T         | GB     | OTH      | MS         | Stock Price Prediction                       |
| [119]                     | 2021 | I    | T         | GB     | OTH      | MS         | Stock Price Prediction                       |
| [60]                      | 2020 | I    | FI        | GB     | GRA      | MS         | Stock Mov. Prediction                        |
| KDTCN [28]                | 2019 | I    | FI        | GB     | GRA      | MS         | Stock Mov. Prediction                        |
| [67]                      | 2022 | I    | FI        | GB     | LR       | MS         | Tick Price Prediction                        |
| $L^3$ [3]                 | 2020 | I    | DR        | GB     | DR       | MS         | Stock Mov. Prediction, Stock Trading         |
| ML-GAT [50]               | 2022 | I    | FI        | GB     | GRA      | MS         | Stock Mov. Prediction                        |
| T2-PALM [35]              | 2022 | I    | DR        | LO, GB | DR       | MS         | Stock Price Prediction                       |
| FRNN [13]                 | 2020 | I    | DR        | LO, GB | FL       | MS         | Stock Price Prediction                       |
| ILSTM [44]                | 2019 | I    | FI        | LO, GB | ATT      | MS         | Stock Price Prediction                       |
| KHTT [63]                 | 2022 | I    | FI        | LO, GB | ATT      | MS         | Stock Mov. Prediction                        |
| FDT-FTS [101]             | 2021 | I    | FI        | LO, GB | DT       | MS         | Stock Price Prediction                       |
| MTNet [20]                | 2018 | I    | FI        | LO, GB | ATT      | MS         | Exchange-Rate Prediction                     |
| MAGNN [22]                | 2022 | I    | FI        | LO, GB | ATT      | MA         | Stock Mov. Prediction                        |
| AT-LSTM [126]             | 2019 | I    | FI        | LO, GB | ATT      | MS         | Stock Price Prediction                       |
| [85]                      | 2020 | I    | FI        | LO, GB | ATT      | MS         |                                              |
| Dandelion [129]           | 2020 | I    | FI        | LO     | ATT      | MS         | Revenue prediction                           |
| AMS [120]                 | 2020 | I    | FI        | LO     | LR       | MS         | Unexpected revenue prediction                |
| IIFI [108]                | 2023 | I    | FI        | GB     | FL       | MS         | Stock Trading, Stock Price Prediction        |
| CTV-TabNet [97]           | 2023 | I    | FI        | LO     | OTH      | MS         | Stock Mov. Prediction, Stock Trading         |
| M-MI [127]                | 2018 | I    | FI        | GB     | LR       | MS         | Stock Mov. Prediction                        |
| [54]                      | 2018 | I    | FI, VE    | LO, GB | OTH      | MS         | Stock Mov. Prediction                        |
| L1-Orth GRU D. Tree [114] | 2020 | I    | DR        | GB     | APP      | MS         | Stock Mov. Prediction                        |
| TABL [104]                | 2019 | I    | FI        | GB     | ATT      | MS         | Stock Mov. Prediction                        |
| [124]                     | 2023 | I    | FI        | LO, GB | OTH      | MS         | Stock Price Prediction                       |
| [36]                      | 2019 | I    | FI        | LO, GB | LR       | MS         | Macroeconomic indicators prediction          |
| [122]                     | 2019 | I    | FI        | LO, GB | ATT      | MS         | Stock Mov. Prediction                        |
| [107]                     | 2020 | I    | FI        | LO     | GRA      | MS         | Stock Mov. Prediction                        |
| [56]                      | 2021 | I    | FI        | LO     | OTH      | MS         | Stock Mov. Prediction                        |
| [61]                      | 2021 | I    | DR        | LO, GB | OTH      | MS         | Stock Mov. Prediction                        |
| [102]                     | 2024 | I    | SA, FI    | LO     | OTH      | MS         | Stock Mov. Prediction                        |
| [49]                      | 2023 | I    | T         | GB     | OTH      |            | Stock Price Prediction                       |

Table 2. Definition of the Abbreviation of Characterization Table

| Acronym | Definition          | Acronym | Definition        |
|---------|---------------------|---------|-------------------|
| E       | Explainability      | I       | Interpretability  |
| FI      | Feature Importance  | CL      | Confidence Level  |
| VE      | Visual Explanation  | T       | Trends            |
| SA      | Sentiment Analysis  | DR      | Decision Rule     |
| GB      | Global              | LO      | Local             |
| APP     | Approximation       | PER     | Perturbation      |
| PRO     | Propagation         | LR      | Linear Regression |
| FIN     | Financial Theory    | DT      | Decision Tree     |
| ATT     | Attention Mechanism | FL      | Fuzzy Logic       |
| GRA     | Graph               | OTH     | Others            |
| MA      | Model Agnostic      | MS      | Model Specific    |

8 Industry Application and Challenges

As shown in Table 1, the two most common financial applications of the reviewed models are stock price prediction and stock movement prediction. This aligns with the primary focus of this survey. The interpretable or explainable models developed for these tasks can be valuable for financial professionals for many reasons. First, and most obviously, because they predict stock behavior, these models assist traders and portfolio managers in deciding when to buy or sell stocks. However, financial professionals rarely rely on a single source of information. Instead, these models serve as an additional analytical tool, complementing sources such as market reports, technical indicators, and fundamental analysis. By integrating predictions from these models into their broader decision-making framework, professionals can refine strategies and reduce uncertainty in transactions.

Furthermore, because these models are interpretable or explainable, they provide more information than traditional black-box models. This transparency allows users to better understand the reasoning behind predictions, fostering trust in the model’s outputs. More importantly, these models help users detect patterns in the data that align with their fundamental analysis. For example, feature importance might highlight relationships between specific market conditions and stock performance, offering financial analysts actionable insights beyond raw predictions.

Additionally, as these models predict stock behavior, they are also applicable to algorithmic trading systems, as demonstrated in Reference [72]. In this context, the model replaces the role of a trader by autonomously managing portfolios based on its predictions. Unlike traditional prediction tasks, where models are evaluated based on error metrics (e.g., RMSE or MAE), algorithmic trading models are assessed using financial performance metrics such as profitability, Sharpe ratios, or maximum drawdown. This shift highlights the practical, outcome-driven nature of these systems.

While explicit examples of market risk assessment were not frequently encountered in the reviewed literature, XAI provides an indirect yet valuable approach to risk analysis. By enhancing the understanding of market behavior, these models enable analysts to associate high-risk periods with shared characteristics. For instance, the feature importance profiles during the 2008 financial crisis might resemble those observed during the COVID-19 market crash in 2020. Even though the causes of these crises differ, their predictive contexts share similarities, particularly regarding indicators of market instability and investor sentiment. Dashboards can consolidate these insights, allowing analysts to identify recurring risk patterns and proactively monitor similar signals in real-time data, potentially identifying emerging risks before they fully materialize.

Despite their usefulness, these models come with certain challenges related to the evaluation of their explainability and interpretability. Comparability between the insights and explanations generated by different models is often difficult, and even when comparisons are attempted, determining a ground truth for explainability remains a significant obstacle. Should these explanations be evaluated against historical financial theories, expert consensus, or specific benchmarks? Additionally, financial theories themselves are not always universally applicable, raising questions about their validity as a reference framework across different market contexts. For example, an explanation considered valid in one economic scenario might lose relevance in another. Another challenge is the inherent noise and ambiguity in financial markets, which can sometimes lead to spurious or misleading explanations rather than meaningful insights. These false patterns may arise from correlations that lack causal relationships, making it difficult to validate the reliability of the extracted explanations. These challenges highlight the need for ongoing refinement, contextual validation, and critical interpretation of the explanations provided by XAI models. Ensuring that these insights remain consistent, actionable, and aligned with real-world financial dynamics is essential for building trust and maximizing their value for financial professionals.

## 9 XAI in Other Fields

This survey primarily focuses on financial time-series prediction, particularly stock market forecasting. However, XAI applications in finance extend beyond time-series analysis to tasks like credit risk assessment, loan approval, fraud detection, portfolio optimization, algorithmic trading, and pricing strategies. By improving transparency and interpretability, XAI promotes trust in AI systems and supports informed decision-making. Beyond finance, healthcare has also emerged as a critical field leveraging XAI to enhance decision-making, medical diagnostics, imaging interpretation, and patient monitoring.

Healthcare utilizes approaches, among others, such as SHAP, LIME, LRP, visual explanation methods, and attention-based models. For instance, [Kwon et al. \[2019\]](#) introduced RetainVis, a tool that combines interpretability with visual analytics for recurrent neural networks applied to electronic medical records. XAI has also been used to identify health patterns: Models leveraging ECG-based features like heart rate variability have distinguished COVID-19 recovered patients from healthy individuals [45], while hybrid approaches combining symptom data and sound analysis have supported early COVID-19 diagnosis [83]. Additionally, interpretable deep learning frameworks have been developed to detect heart anomalies [32], highlighting XAI's role in advancing clinical applications.

While both healthcare and finance may face a choice between interpretability and accuracy, their adoption and validation processes differ. Healthcare often prioritizes explainable models due to the complexity of medical data and the high stakes of clinical decisions [95]. However, as noted by [31], explanations are inherently more abstract and distanced from the model compared to direct interpretability. Rigorous evaluation is essential in both fields, but healthcare benefits from clinical validation, where models must align with established medical knowledge to be utilized. In finance, validation is often theoretical or less definitive, reflecting the abstract nature of financial data. Additionally, healthcare AI must adhere to strict ethical and legal standards to ensure unbiased, safe, and reliable decisions, including fairness across patient demographics such as gender and ethnicity. While fairness is also relevant in finance, its impact is less pronounced due to differences in the nature of the data.

XAI's ability to enhance transparency benefits not only professionals but also nonexpert stakeholders. In healthcare, this helps patients understand medical diagnoses and treatment options, while in finance, portfolio managers can explain investment decisions to clients. However, the stringent standards in healthcare often delay the adoption of XAI compared to the relatively

flexible deployment seen in finance. This highlights the need for further refinement of XAI models in healthcare to meet these demands.

However, in the legal field, explainability is crucial due to the necessity of providing justifications for decisions, often requiring symbolic reasoning or case-based approaches. Unlike finance, where the primary concern is ensuring that predictions are reliable for decision-making, legal AI must conform to normative and procedural constraints that demand explicit, human-understandable reasoning. This results in the use of models based on symbolic logic and machine learning, requiring explanations tailored to legal professionals and citizens. Moreover, there is a strong divergence in explanation methods depending on the legal subfield (e.g., administrative law, criminal law), where different evidentiary thresholds and procedural rules shape how AI-generated decisions should be justified [92]. This divergence underscores why certain models may be effective in finance but require additional layers of interpretability in legal applications.

## 10 Conclusion and Future Works

In this survey, the most recent interpretable models and explainability methods applied to financial time-series prediction are presented. This article can serve as a useful reference for selecting XAI approaches in finance, as it is specifically composed to assist users in this application. We first outline the XAI problems and provide definitions of some most important concepts established by the community. Then, we introduce two taxonomies to guide readers in selecting the appropriate models based on their specific needs. We also highlight applications of XAI in the financial industry and outline the challenges of XAI in finance. Finally, we compare XAI in finance with its applications in healthcare and the legal field. According to the surveyed publications, feature importance is clearly the preferred XAI principle in financial time-series prediction. This preference makes sense, because most often users want to know how a model weights its input, like the importance a portfolio manager would naturally give to financial variables. It is also more accessible than the other principles.

There appears to be a greater focus on developing interpretable models as compared to explainable models. This trend could be due to the context dependency of the financial models. Indeed, models used for financial time-series prediction are often specifically designed for this task. The designers of these models might want to develop models that could be understandable by humans, namely XAI models, to apply them in practice. It seems logical to create a model with some interpretable components from the beginning, rather than creating a black-box model and explaining it later. In fact, very often, the concepts and the techniques of explainability methods could be directly applied to AI models to make them interpretable or to improve their interpretability and accuracy. Nothing prevents explaining the interpretable model later using an explainability method if the interpretations are insufficient.

Based on this review, we can say that linear regression, attention mechanism and ruled based models are the most popular techniques for making an interpretable model, while the SHAP algorithm is the most prevalent among explainability methods. The popularity of SHAP is due to its robust mathematical foundation and ease of use.

We emphasize that explainability and interpretability are distinct concepts that should be treated separately due to their unique applications and utilization. According to References [59, 94], interpretable models should be preferred over explainability methods applied to black-box models in high-risk decision-making. For instance, Rudin [2019] strongly argues against the necessity of a tradeoff between accuracy and interpretability in model design, as supported by examples in References [21, 85]. Additionally, explanations from an explainability method are not always faithful to the model and may lack accuracy or completeness. In contrast, the interpretable principles of a model are, by definition, faithful to it.

Interpretable models, however, are not without challenges. According to Reference [94], training these models can be computationally demanding due to their complex architecture, often requiring more effort to optimize for a specific task. Unlike black-box models, which can sometimes learn unexpected patterns with minimal adjustment, interpretable models typically require domain knowledge to be designed effectively. Developing an interpretable model that matches the accuracy of a black-box model often takes more time and effort. This can create a tradeoff between interpretability and accuracy, as suggested in Reference [8], which may hold in certain cases. Oversimplifying a model in pursuit of interpretability or misadjusting it can lead to a loss of crucial information, potentially affecting decision-making accuracy. In financial contexts, this may influence investment strategies, asset valuation, or credit assessments, with potential repercussions on overall financial stability and market exposure. Additionally, as discussed in Section 5.4, the reliability of interpretations for complex models remains an open question. In scenarios where a well-performing model is already in use, applying explainability methods to decipher it may sometimes be more reasonable than rebuilding an interpretable alternative.

We would like to finish the article by some propositions for future works. Few studies, if any, rigorously tested the faithfulness and the reliability of their explainability methods; a ranking based on explainability levels could be enlightening. Moreover, the absence of a quantitative metric for model interpretability suggests the need for further work in this domain. Additionally, since most XAI models have not been tested with the same benchmarks regarding the performance, the interpretability/explainability, the selection of the best approach remains challenging. More quantitative studies will be necessary for the development and the application of the XAI approaches. As the trend continues to move toward causal XAI and federated learning, further research could focus on integrating multiple XAI principles within a single model to maximize the information extracted. For instance, combining feature importance with decision rules could significantly enhance users' understanding of both the process and the model's outputs. Additionally, developing hybrid models that balance high predictive power with strong interpretability remains a promising research direction. Finally, improving methods for quantifying model uncertainty would facilitate more informed model selection and increase trust in AI-driven decisions.

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